

Rudraksh Fanse

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EDUCATION

Sardar Vallabhbhai National Institute of Technology

B.Tech in Computer Science | CGPA: 9.51

Surat, Gujarat

2024 – Present

EXPERIENCE

Research Intern — Computational Science

Summer Intern 2026

Indian Institute of Technology Gandhinagar (IIT Gandhinagar)

On-Site

- Simulated reaction-diffusion dynamics across multiple canonical systems to build a labeled dataset of thousands of Turing and non-Turing patterns for machine learning classification.
- Derived pattern-formation instability thresholds analytically and validated dataset labels against theory, filtering out physically inconsistent samples near critical boundaries.
- Automated large-scale data generation through parameter sweeps over diffusion coefficients, producing thousands of simulations spanning distinct pattern regimes.
- Engineered an FFT-based feature-extraction pipeline and trained a classifier to distinguish Turing from non-Turing patterns.

PROJECTS

Joy Juncture — Gamified E-Commerce Platform | *Next.js, TypeScript, Firebase, Redis, Razorpay*

2026

- Built a full-stack gamified commerce platform with 14+ games, a 4-tier XP engine, and 30+ APIs.
- Implemented Redis-based leaderboards and analytics achieving 10× query speedup.
- Integrated Razorpay with secure payment verification and automated rewards.

Enfinity Vision Pro — Offroad Image Segmentation | *Next.js, React, Python, Hugging Face, Gradio*

2026

- Built a full-stack computer vision platform for high-precision offroad image segmentation, integrating a custom-trained Hugging Face model via the Gradio SSE v3/v5 streaming protocol.
- Engineered a hardened server-side inference API with 600s timeout support using undici, enabling long-running TTA (Test Time Augmentation) tasks without connection drops.
- Implemented a multi-view result explorer with real-time toggle between Input, Colorized Mask, and Final Composite Overlay outputs with live inference status feedback.

Genesis — AI Multi-Agent Content Platform | *Next.js, FastAPI, LangGraph, PostgreSQL, Redis*

2026

- Built a multi-agent AI workflow (Coordinator → Planner → Executor → Reviewer) using LangGraph for structured, stateful content generation.
- Integrated Gemini 2.5 Flash via LangChain and implemented RAG using pgvector (1408-dim embeddings) with cosine similarity search.
- Designed a 3-layer caching system (Redis → Supabase → PostgreSQL) reducing redundant LLM calls, with OAuth 2.0 publishing pipelines for LinkedIn and Twitter/X.

TECHNICAL SKILLS

Programming: C++, Python, JavaScript, C, HTML/CSS

Web & Backend: React, Next.js, Node.js, Express.js

Databases & Tools: MongoDB, Firebase, Git, GitHub

ACHIEVEMENTS

Won Genesis — Generative AI Competition (Blog/Content Creation Track).

Won GWOC (Google Winter of Code) — Built website for Joy Juncture.

Won Enfinity Hackathon — Built DL model for Semantic Segmentation of Desert Images.

POSITIONS OF RESPONSIBILITY

Treasurer — ACM, SVNIT.

Junior Developer — GDG, SVNIT.

Executive — CEV, SVNIT.